TASK 5: INDIVIDUAL CASE SAFETY REPORT (ICSR)

1. Worldwide Unique Case Identification

Regulatory Authority / Organisations	Indian Pharmacopoeia Commission (PvPI)
Worldwide Unique Case ID	-
Receipt Date	29/09/2026
Type of Report	Spontaneous

2. Patient Characteristics

Patient Initials / Identifier	PHCG
Age at Time of Event	92 years
Gender	Male
Weight	-
Allergy Status	No known allergy
Medical History	Alzheimer’ s disease, old ischaemic heart disease (IHD), and benign prostatic hyperplasia (BPH)
Relevant Clinical Conditions	Acute cerebrovascular accident (CVA) with intracranial bleed extending to the right corona radiata and left hemiparesis; sepsis due to aspiration pneumonia (resolving)

3. Adverse Drug Reaction(s)

Suspected Reaction	Hypokalemia (Decreased Serum Potassium)
Reaction Start Date	24/09/2026
Reaction Stop / Resolution Date	27/09/2026
Serum Potassium Trend	23/09/2026: 3.6 mmol/L; 24/09/2026: 3.2 mmol/L; 25/09/2026: 3.2 mmol/L; 26/09/2026: 3.0 mmol/L; 27/09/2026: 3.6 mmol/L.
Outcome	Recovered
Seriousness	Not Serious
Expectedness	Expected

4. Drug Information — Suspected Drug

Drug Role	Suspected
Medicinal Product Name	Inj. Mannitol
Active Ingredient	Mannitol
Dose / Route / Frequency	100 mL, Intravenous (IV), Once daily (OD),
Indication	Acute CVA
Therapy Start Date / Stop Date	22/09/2026 to 25/09/2026
Action Taken with Drug	Drug withdrawn
Causality	Probable

5. Concomitant / Corrective Treatment

Syrup Potklor	10 ml Three Times Daily (TID), started on 24/09/2026
Inj. KCl	2 Ampoules in 500 mL IV fluids at 100 mL/hour, given on 25/09/2026
Other Concomitant Treatment	Neb Duolin given on 19/09/2026 to 10/10/2026 for Aspiration Pneumonia/Sepsis.

6. Relevant Diagnostic Tests & Laboratory Data

Serum Potassium	23/09/2026: 3.6 mmol/L
Serum Potassium	24/09/2026: 3.2 mmol/L
Serum Potassium	25/09/2026: 3.2 mmol/L
Serum Potassium	26/09/2026: 3.0 mmol/L
Serum Potassium	27/09/2026: 3.6 mmol/L
Blood Pressure	110/70 mmHg
Heart Rate	120 Beats/Min
Respiratory Rate	22 Breaths/Min

7. Case Narrative

PHCG is a 92-year-old male with Alzheimer’s disease, Old IHD, and BPH who was admitted with Acute CVA involving an Intracranial Bleed Extending to The Right Corona Radiata with Left Hemiparesis, along with Sepsis due to Aspiration Pneumonia (resolving). Intravenous mannitol as the suspected drug, recorded as 100 mL IV once daily from 22/09/2026 to 25/09/2026 for Acute CVA. Serum Potassium was 3.6 mmol/L on 23/09/2026, decreased to 3.2 mmol/L on 24/09/2026 and remained 3.2 mmol/L on 25/09/2026, then decreased to 3.0 mmol/L on 26/09/2026. Syrup Potklor 10 mL TID was started on 24/09/2026, and 2 ampoules of IV KCl in 500 mL fluids at 100 mL/hour were given on 25/09/2026. Potassium was 3.6 mmol/L on 27/09/2026 and the Outcome was recorded as Recovered. Inj Mannitol was withdrawn and records Causality as Probable. Mannitol-induced osmotic diuresis may Increase Urinary Potassium Loss; however, acute illness, sepsis, nutritional status, and other treatments may also contribute. The causality category is reported as Probable and should be interpreted alongside the treating assessment.

8. Pharmacological Rationale

Mechanism	Mannitol increases osmotic pressure in the renal tubules, reducing water reabsorption and increasing urine flow. Increased tubular flow and distal sodium delivery can promote potassium secretion into urine, potentially Decreasing Serum Potassium.
Mechanism Summary	Inj. mannitol → increases tubular osmotic pressure → reduces water reabsorption and increases urine output → increases distal sodium delivery/tubular flow → promotes urinary K^+ loss → hypokalemia.

9. ICH E2A / Case Assessment Summary

Seriousness	No Seriousness
Expectedness	Expected
Causality Assessment	Probable
Outcome	Recovered by 27/09/2026, with serum potassium recorded as 3.6 mmol/L

THIS WAS DONE BASED ON CASE SCENARIO

CASE SCENARIO:

Patient Name: PHCG

Age: 92 Year

Gender: Male

Allergy Status: No Known Allergy

Past Medical History: Alzheimer’s Disease, Old IHD, BPH

Present Illness: Acute CVA – GC bleed extending to right corona radiata with left hemiparesis, Sepsis – Aspiration Pneumonia (Resolving).

Past Medication History:

T. ESCOSPRIN AV (75/10MG) 0-0-1
T. VELTAM S 0-0-1
T. DAPAGLIFLOZIN 5MG 0-1-0
T. CIDMUS 50MG 1/2-0-0
T. CARDIVAS CR 10MG 1-0-0
T. OXRA 5 MG 1-0-0

On 24/09/2026 The Patient Serum Potassium Level Declined to **3.2 mmol/L**

On 25/09/2026 The Patient Serum Potassium Level Declined to **3.2 mmol/L**

On 26/09/2026 The Patient Serum Potassium Level Declined to **3.0 mmol/L**

LABORATORY DATA:

Physical Examination:

Blood Pressure: 110/70 mmHg

Heart Rate: 120 BPM

Respiratory Rate: 22CPM

MANAGEMENT:

On 24/09/2026 She was started with Syp Potklor 10ml TID

On 25/09/2026 She was treated with Inj KCl 2 AMP in 500ml fluids @ 100CC/hrs

The Decreased Serum Potassium Level (Hypokalemia) was resolved on 27/09/2026

ADR REPORT FORM OF THIS CASE SCENARIO

Version 1.4

SUSPECTED ADVERSE DRUG REACTION REPORTING FORM

For **VOLUNTARY** reporting of ADRs by Healthcare Professionals

INDIAN PHARMACOPOEIA COMMISSION (National Coordination Centre-Pharmacovigilance Programme of India)

Ministry of Health & Family Welfare, Government of India, Sector-23, Raj Nagar, Ghaziabad-201002

PvPI Helpline (Toll Free) :1800-180-3024 (9:00 AM to 5:30 PM, Monday-Friday)

Initial Case ☑ | Follow-up Case ☐

FOR AMC / NCC USE ONLY

Reg. No. / IPD No. / OPD No. / CR No. : 2200086009

AMC Report No. :

Worldwide Unique No. :

A. PATIENT INFORMATION *

1. Patient Initials: PMCH
2. Age or date of birth: 90 Year
3. Gender: M ☑ F ☐ Other ☐
4. Weight (in Kg.)

B. SUSPECTED ADVERSE REACTION *

5. Event / Reaction start date (dd/mm/yyyy): 24-09-2026
6. Event / Reaction stop date (dd/mm/yyyy): 27-09-2026
7. Describe Event/Reaction management with details , if any

A Patient was treated with Inj Mannitol for Acute CVA for which the patient developed Decreased Serum Potassium level. (Hypokalemia).

12. Relevant investigations with dates : Sr K+ level :
23-09-2026 - 3.6 mmol/L 27-09-2026 - 3.6 mmol/L
24-09-2026 - 3.2 mmol/L
25-09-2026 - 3.2 mmol/L
26-09-2026 - 3.0 mmol/L

13. Relevant medical / medication history (e.g. allergies, pregnancy, addiction, hepatic, renal dysfunction etc.)
K/C/O : Alzheimer's disease, Old IHD, BPH.
Diag : Acute CVA - GC bleed extending to right corona radiata with left hemiparesis, Sepsis - Aspiration Pneumonia.

14. Seriousness of the reaction : No ☑ If Yes ☐ (*please tick anyone*)
☐Death (dd/mm/yyyy) ☐Congenital-anomaly
☐Life threatening ☐Disability
☐Hospitalization-Initial/Prolonged ☐Other Medically important

15. Outcome:
☑Recovered ☐Recovering ☐Not Recovered
☐Fatal ☐Recovered with sequelae ☐Unknown

C. SUSPECTED MEDICATION(S) *

S. No.	B. Name (Brand/ Generic)	Manufacturer (if known)	Batch No. / Lot No.	Expiry Date (if known)	Dose	Route	Frequency	Therapy Dates: Date Started	Therapy Dates: Date Stopped	Indication	Causality Assessment
i	Inj Mannitol (Mannitol)				100ml	IV	BD	24-09-	25-09	Acute CVA	Probable
ii											
iii											
iv#											

9. Action taken after reaction (*please tick*)

S. No. as per C	Drug withdrawn	Dose increased	Dose reduced	Dose not changed	Not applicable	Unknown
i	✓					
ii						
iii						
iv						

10. Reaction reappeared after reintroduction of suspected medication (*please tick*)

Yes	No	Effect unknown	Dose (if re-introduced)

11. Concomitant medical product including self-medication and herbal remedies with therapy dates (Exclude those used to treat reaction)

S. No.	Name (Brand / Generic)	Dose	Route	Frequency (OD, BD, etc.)	Therapy Dates: Date Started	Therapy Dates: Date Stopped	Indication
i	Neb Duolin 1.25mg + Levosalbutamol 500mcg + Ipratropium)	1.25mg +	Mask	Q8hr	19-09-2026	10-10-2026	Sepsis - Aspiration Pneumonia.
ii							
iii#							

Additional Information :

On 24-09-2026 - Syp Potklon 10ml TID given.

On 25-09-2026 - Inj KCl @ Amp in 500ml fluids @ 100cc/hr.

D. REPORTER DETAILS *

16. Name & Address : Harshitha
FMMCH

Pin : 535098 Email : acharyaharshitha38@gmail.com

Contact No- : 8867203149

Occupation : Pharm D Intern Signature : Harshitha

17. Date of this report (dd/mm/yyyy) : 29-09-2026

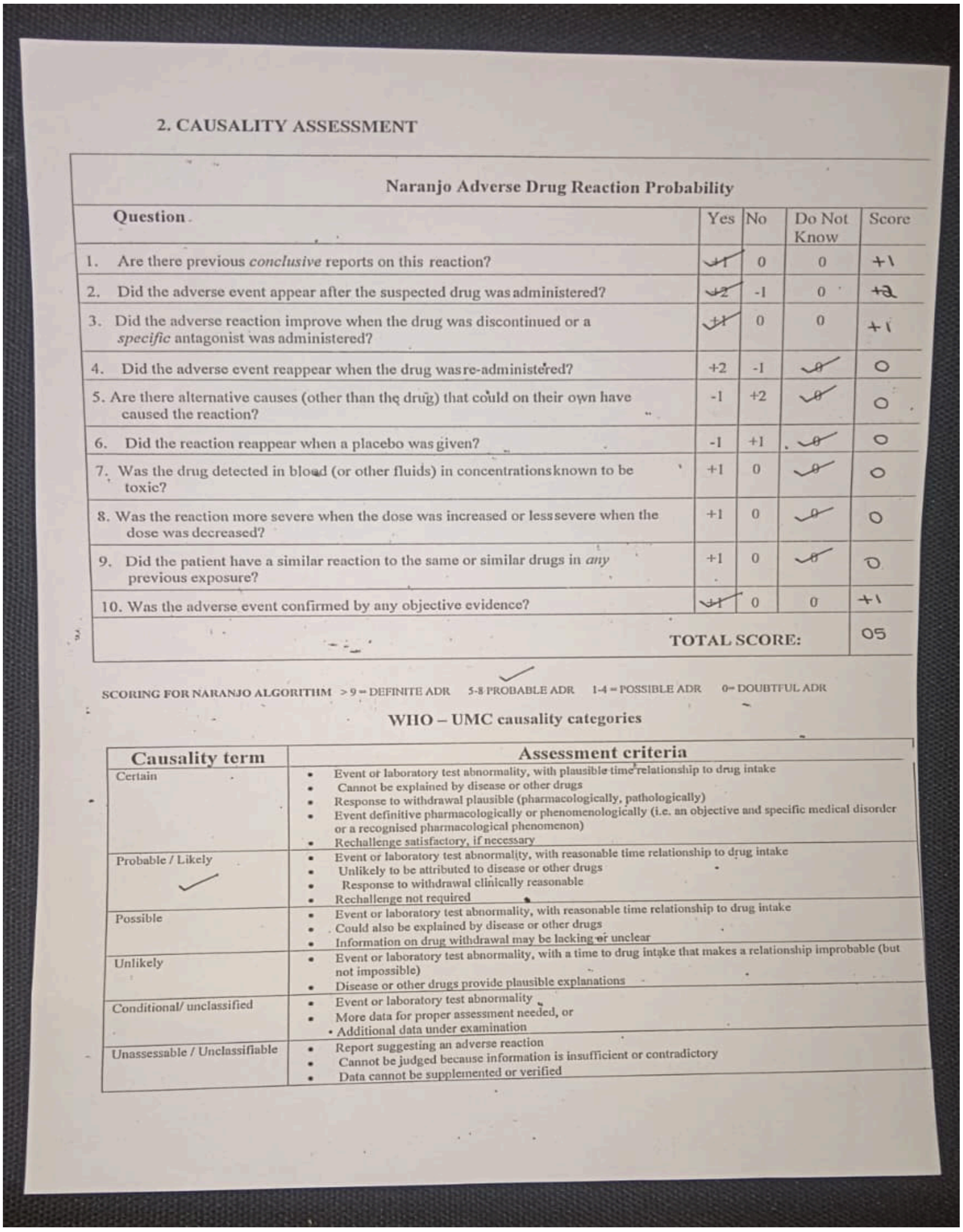
Signature and Name of Receiving Personnel :

Confidentiality : The patient's identity is held in strict confidence and protected to the fullest extent. Submission of a report does not constitute an admission that medical personnel or manufacturer or the product caused or contributed to the reaction. Submission of an ADR report does not have any legal implication on the reporter.

Use separate page for more information

* Mandatory Fields for suspected ADR Reporting Form

CAUSALITY ASSESSMENT USING NARANJO SCALE AND WHO-UMC SCALE OF CASE SCENARIO



2. CAUSALITY ASSESSMENT

Naranjo Adverse Drug Reaction Probability

Question	Yes	No	Do Not Know	Score
1. Are there previous *conclusive* reports on this reaction?	+1	0	0	+1
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0	+2
3. Did the adverse reaction improve when the drug was discontinued or a *specific* antagonist was administered?	+1	0	0	+1
4. Did the adverse event reappear when the drug was re-administered?	+2	-1	0	0
5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	-1	+2	0	0
6. Did the reaction reappear when a placebo was given?	-1	+1	0	0
7. Was the drug detected in blood (or other fluids) in concentrations known to be toxic?	+1	0	0	0
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	0
9. Did the patient have a similar reaction to the same or similar drugs in *any* previous exposure?	+1	0	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	+1
			TOTAL SCORE:	05

SCORING FOR NARANJO ALGORITHM > 9 = DEFINITE ADR 5-8 PROBABLE ADR 1-4 = POSSIBLE ADR 0= DOUBTFUL ADR

WHO – UMC causality categories

Causality term	Assessment criteria
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e. an objective and specific medical disorder or a recognised pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable / Likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations
Conditional/ unclassified	• Event or laboratory test abnormality • More data for proper assessment needed, or • Additional data under examination
Unassessable / Unclassifiable	• Report suggesting an adverse reaction • Cannot be judged because information is insufficient or contradictory • Data cannot be supplemented or verified